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Electronic Road Pricing: Developments after phase I

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The [Land Transport Authority](#) (LTA) launched the Electronic Road Pricing (ERP) system in April 1998 as a new way to manage traffic congestion. Phase I was completed in September 1998 with the activation of ERP for [East Coast Parkway](#) (ECP), [Central Expressway](#) (CTE), [Pan-Island Expressway](#) (PIE) and the city area. The number of ERP gantries in operation has since increased to 71, up from 33 at the end of 1998.¹ Improvements have also been made to the scheme since it was first introduced, including credit card payment of ERP charges and graduated fee pricing.²

Expansion of ERP Coverage

Since 1998, more expressways have been covered under the ERP system. These are the Ayer Rajah Expressway (AYE), Bukit Timah Expressway (BTE), [Kallang-Paya Lebar Expressway](#) (KPE) and Marina Coastal Expressway (MCE).³ The ERP network has also extended to arterial roads outside the city in response to the heavy city-bound traffic on these roads. The first few roads

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Since 2005, evening ERP has been applicable for the ECP, MCE, AYE and the central business district (CBD) to target the evening peak hour traffic.⁶

Major enhancements

Graduated fee pricing

After observing that motorists often sped up just before the start of a time period with higher ERP rates, or slowed down just before lower rates kicked in, LTA introduced a graduated fee pricing in 2003. The objective was to reduce the incentive for such behaviour by moderating the sharp changes in rates between two successive time periods. This was done by implementing five-minute buffer periods during which the ERP rate would be midway between the rates of the two successive periods.⁷

85th-percentile speed measurement method

Every quarter, LTA reviews the traffic conditions on roads where the ERP system is in operation. After the review, it adjusts the rates where necessary so as to achieve optimal traffic flow on these roads.⁸ It also monitors other roads closely to identify new locations where congestion levels warrant ERP. When LTA first introduced the system, it said that ERP would be activated or the charges at existing gantries would be raised when average speeds fell below the threshold levels of 45 kmh for expressways and 20 kmh for arterial roads. However, the use of average speeds to determine when ERP rate changes were needed could create "stop-start" conditions on the roads easily. Thus, since July 2008, LTA has used the 85th percentile speed method to measure actual traffic conditions. With this, "motorists will be assured of smooth travel on ERP-priced roads at least 85 percent of the time".⁹

Orchard Road cordon

As the Orchard Road area is primarily a shopping destination, its traffic pattern differs from that of the rest of the city centre. The road experiences heavier traffic later in the day, after the retail outlets have opened. To better

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Improved in-vehicle unit

LTA has introduced a second-generation in-vehicle unit (IU) that offers several improvements over the old model. For example, the new IU can accept both contact and contactless stored-value cards, whereas the previous model accepted only contact cards, specifically the NETS CashCard. Another new feature is that it can automatically top up a card that registers insufficient funds when the vehicle passes through an ERP gantry. The improved IU was rolled out to new motorcycles and installed in all other new vehicles from 2009 onwards. Owners of existing vehicles can purchase the new device or continue to use the earlier model.¹¹

From 2008, cardless ERP payment systems, such as EZ-Pay, MotorPay and vCashCard, were introduced to allow motorists to pass through ERP gantries without a CashCard in their IU. With these systems, all the charges are billed to a registered credit card.¹²

Timeline

Sep 1999: ERP extended to AYE and arterial roads outside the city.

Feb 2003: Graduated fee pricing introduced.

Sep 2003: New scheme launched for foreign-registered cars to give added convenience to foreign travellers – instead of installing or renting an IU, motorists can opt to pay a fixed daily ERP fee.¹³

Aug 2005: Evening ERP introduced for home-bound traffic leaving the city.

Oct 2005: Separate ERP cordon for Orchard Road area is implemented.

Oct 2007: KPE partially opened, with one new ERP gantry activated at the south-bound KPE exit into west-bound ECP.¹⁴

Nov 2007: ERP kicks in on BKE.¹⁵

Mar 2008: Government announces that new signboards, known as the Rates Variable Message System, will be installed at the top of gantries to prominently display the prevailing charges for various vehicle types. This is an

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commences.

Sep 2008: KPE is fully opened, with a total of 16 ERP gantries installed, the largest number of gantries on a single road. However, the gantries would only be activated when congestion builds up and travel speeds fall below 45kmh.¹⁸

Nov 2008: New scheme introduced to allow ERP charges incurred by motorists to be billed to their credit cards, as an alternative to having the charges deducted from the CashCard in the IU.

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